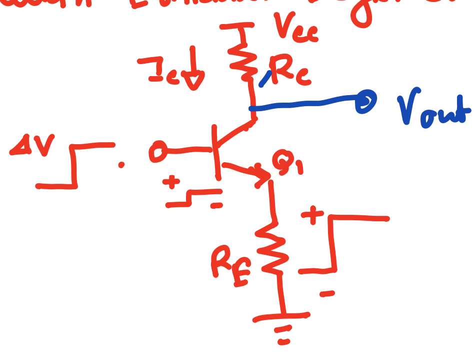
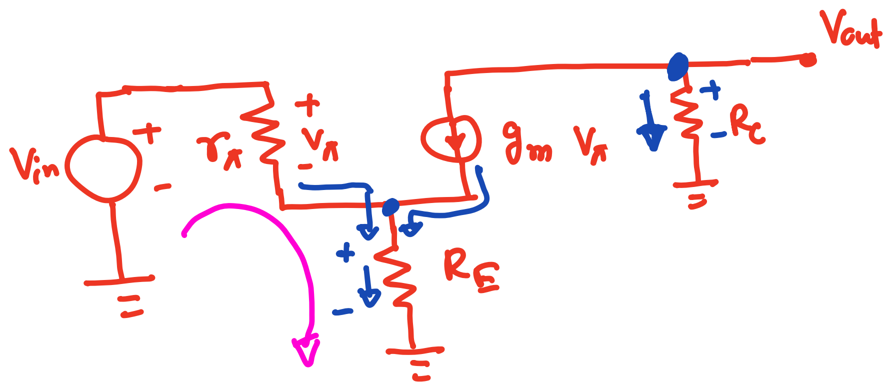
CE stage with Emitter Degeneration



$I_C = I_S \exp \frac{\Delta V_{BE}}{V_T}$



$A_V = \frac{V_{out}}{V_{in}}$

KCL @ output node

$$g_m V_\pi + \frac{V_{out}}{R_C} = 0$$

$$\therefore g_m V_\pi = -\frac{V_{out}}{R_C}$$

$$\therefore V_\pi = -\frac{V_{out}}{g_m R_C} \quad \text{—— (1)}$$

$$V_{RE} = \left(\frac{V_\pi}{r_\pi} + g_m V_\pi\right) R_E \quad \text{—— (2)}$$

$$V_{in} = V_\pi + V_{RE}$$

$$= V_\pi + \left(\frac{V_\pi}{r_\pi} + g_m V_\pi\right) R_E \quad \text{[using eq 2]}$$

$$= V_\pi \left[1 + \left(\frac{1}{r_\pi} + g_m \right) \right] R_E$$

using eqⁿ 1

$$\Rightarrow V_{in} = -\frac{V_{out}}{g_m R_C} \left[1 + \left(\frac{1}{r_\pi} + g_m \right) \right] R_E$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = -\frac{g_m R_C}{1 + \left(\frac{1}{r_\pi} + g_m \right) R_E}$$

$\beta \gg 1, \quad g_m \gg 1/r_\pi$

$$\therefore \frac{V_{out}}{V_{in}} = \frac{-g_m R_C}{1 + g_m R_E}$$

$$= \frac{-\cancel{g_m} R_C}{\cancel{g_m}(1/g_m + R_E)}$$

$$\boxed{\therefore \frac{V_{out}}{V_{in}} = -\frac{R_C}{\frac{1}{g_m} + R_E}}$$

without degeneration $R_E = 0$

$$\therefore \frac{V_{out}}{V_{in}} = -g_m R_C$$

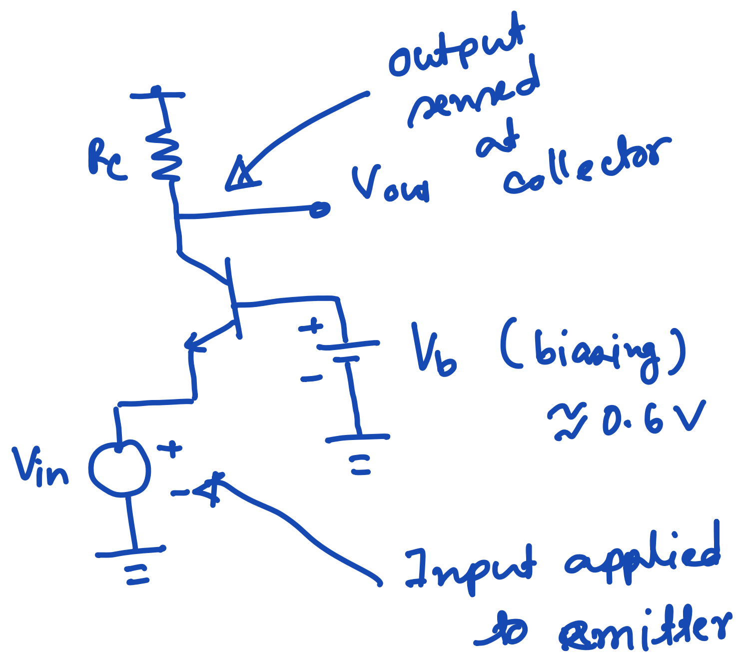
Example 5.22, 5.23

Input Impedance:

$$R_{in} = r_\pi + (1+\beta) R_E$$

Output Impedance: $R_{out} = R_C$

Common-Base Topology



Voltage gain:

$$A_V = g_m R_C$$

$$g_m = \frac{I_C}{V_T}$$

Input impedance:

$$R_{in} = \frac{1}{g_m}$$

Output Impedance:

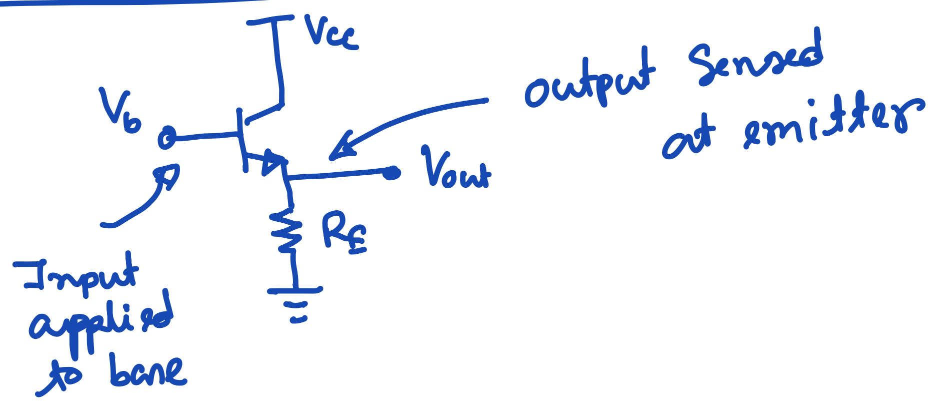
$$R_{out} = R_C \quad \text{if } V_A = \infty$$

$$\text{or } R_{out} = R_C \| r_o \quad \text{if } V_A \neq \infty$$

Example 5.42 (Tough problem)

10 marks.

Common collector (or emitter follower)



$$A_v = \frac{R_E}{R_E + \frac{1}{g_m}}$$

$$R_E \gg \frac{1}{g_m}$$

$$A_v < 1$$

Example 5.43, Examples 5.45